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A 2D fully coupled hydro-mechanical finite-discrete element model with real pore seepage for simulating the deformation and fracture of porous medium driven by fluid



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ABSTRACT

Based on the finite-discrete element method (FDEM), a 2D fully coupled model with real pore seepage is proposed. This model can solve the problem of the deformation and fracture of porous medium driven by fluid. In this model, the fluid flow in the fracture is expressed by the cubic law, while the fluid flow in the rock matrix is characterized by Darcy's law and solved by the finite volume method. The interaction between pore seepage and fracture seepage is realized at the fracture. Three analytical solutions are presented to verify the correctness of the proposed model. The results show that the numerical solutions agree well with the analytical solutions. In addition, a hydraulic fracturing problem with a complex fracture network is studied using this model. The simulation results show that the model can capture the fracture initiation, propagation, and intersection, the interaction of natural fractures and newly generated fractures, and the evolution of fluid pressure during hydraulic fracturing. The model can be used not only to simulate hydraulic fracturing in shale gas and geothermal mining but also to solve a series of geomechanical problems related to the effect of fluid. Thus, this model has broad application prospects.

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1. Introduction

Hydro-mechanical coupling refers to the interaction of seepage and mechanics. Many geomechanical problems are related to hydro-mechanical coupling, such as landslide or slope failure, dam foundation damage, stability of excavation, design and safety assessment of nuclear waste underground disposal, oil and gas exploration and geothermal mining, hydraulic fracturing, compression and settlement of reservoirs, hydraulic fracturing, coal mining, coalbed methane extraction, underground storage of fluids, oil and gas geological storage, and toxic liquid waste disposal.

1.1. Analytical methods for hydro-mechanical coupling

Classic models of hydro-mechanical coupling of porous media such as soil can be traced to the effective stress principle and the one-dimensional theory of consolidation proposed by Terzaghi [1]. Extending the work of Terzaghi, Biot [2,3] used the theory of elasticity and Darcy's law to describe solid behavior and interstitial fluid flow, respectively, denoted Biot consolidation theory. Based

on this theory, several enhanced hydro-mechanical coupling formulations have subsequently been proposed to study the pore mechanical response of fractured-porous rock masses [4–6].

Another important driving force for the study of hydro-mechanical coupling in rock is the extensive application of hydraulic fracturing technology in oil and shale gas exploitation. A series of analytical models of hydraulic fracturing have been established, such as the PKN model [7,8], KGD model [9,10], the radial or penny-shaped model [11], and some pseudo-three dimensional (P3D) and three-dimensional (PL3D) models [12]. Recent progress in the analytical modeling of hydraulic fracturing has been reviewed elsewhere [13].

1.2. Numerical methods of hydro-mechanical

Analytical methods of hydro-mechanical coupling are often oversimplified and generally consider a homogenous elastic medium or simple geometry. Experimental methods are not only expensive but also extremely tedious and time consuming. Thus, experimental methods are not efficient, especially for routine industry application. Therefore, numerical methods have been established for hydro-mechanical coupling. For example, some continuum numerical methods have been adopted for the simulation of hydraulic fracturing, of which the most representative are

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the displacement discontinuity method (DDM) [14–16], the finite element method (FEM) [17–22], and the extended finite element method (XFEM) [23,24]. However, these methods have some limitations in simulating hydraulic fracturing. For example, in DDM, fluid can flow through closed natural fractures, but the rock itself is still impermeable. Thus, DDM cannot consider fluid loss into the rock matrix. A drawback of FEM in the simulation of hydraulic fracturing is that the modeling of crack growth requires grid remeshing, which is computationally burdensome and involves the transfer of data between different meshes. Although XFEM avoids mesh regeneration, it requires a very fine mesh in the solution domain, which leads to a huge computational load when the crack propagation path is unknown in advance. Thus, it is difficult to simulate the propagation and intersection of complex multiple cracks using XFEM. Moreover, a common problem of the continuum numerical methods is that they cannot adequately handle the contact when the crack is closed.

Discontinuous numerical methods are another tool for simulating hydraulic fracturing that have been used widely in recent years. For example, discrete element methods (such as UDEC/3DEC) [25,26] can simulate crack initiation and propagation by introducing a virtual joint between the divided sub-blocks. Thus, combined with hydro-mechanical, discrete element method can be used to simulate hydraulic fracturing [27–29]. Similarly, the discontinuous deformation analysis method can also be used for simulating hydraulic fracturing by introducing the same technology [30–34]. The numerical manifold method is another potential method. The coupled hydro-mechanical model in Ref. [42] is also used in the numerical manifold method [62], which makes it possible to simulate hydraulic fracturing. However, these discontinuous numerical methods all consider the fluid flow in fractures but do not take into account fluid leakage from fractures into the surrounding rock matrix; i.e., the pore seepage into the rock matrix is neglected. The particle discrete elements method (PFC) [35] bonds a series of balls together to characterize the continuum. By breaking inter-spherical bonds, the method simulates the fracture and fragmentation of the solid, which can be used to simulate hydraulic fracturing combined with a hydro-mechanical coupling module [36–40]. However, the microscopic parameters of the method are difficult to calibrate, and the fracture characterization is not intuitive. In addition, the concept of stress and strain in continuum mechanics no longer directly exists in this method.

To overcome the defects of the above methods in simulating hydraulic fracturing, this paper proposes a two-dimensional fully coupled model for simulating the deformation and fracture driven by fluid is proposed based on the finite discrete element method (FDEM). The model is presently implemented in the Y-flow code [41,42]. The advantage of this model is that it takes into account both the fracture seepage in the fracture and the real pore seepage in the rock matrix, in contrast to the equivalent method [43–45]. Thus, the model considers the permeability anisotropy of the rock matrix and the fluid loss from the fracture into the rock matrix. Furthermore, the model can simulate the crack initiation and propagation, the interaction of complex fractures, and the interaction between fluid and solid. Since cracks extend along the element boundary, remeshing is not needed. The cracks consist of line segments, and the characterization of the cracks is very intuitive. FEM is used to calculate the stress and strain of a triangular element, and thus the concept of stress and strain in continuum mechanics is well preserved in FDEM. In addition, the discrete element method is used to process contacts between elements in FDEM; the contacts can be easily handled when the crack is closed.

The paper is organized as follows. First, the fundamentals of FDEM are briefly introduced. In Section 3, the basic assumptions of the fully coupled hydro-mechanical model are introduced. The

2D coupled model with real pore seepage and fracture seepage is introduced in detail in Section 4, which introduces the basic idea of hydro-mechanical coupling, the fracture seepage flow model, the two-dimensional pore seepage model, the coordination process between fracture seepage and pore seepage at the fissure, the application of fluid pressure in fractures and pore pressure in the rock matrix, and the fracture-stress-seepage coupling process. In Section 5, the model is verified thoroughly using three examples with analytical solutions to test the effectiveness of the model in addressing unsteady saturated seepage, unsteady unsaturated seepage and hydro-mechanical coupling in porous media. The fifth example is a hydraulic fracturing problem with complex fracture networks. The interaction of hydraulic fractures and natural fractures is studied.

2. The fundamentals of the finite discrete element method (FDEM)

FDEM is a general numerical method for simulating the fracturing of solid materials [41,46–48]. It combines the advantages of FEM for the simulation of solid deformation and the advantages of the discrete element method to address contacts between blocks. In addition, non-linear fracture mechanics are introduced in the joint element. This method has become widely used in the field of rock mechanics, especially for problems related to rock fracture [42,49–54].

The fundamental basis of two-dimensional FDEM is that the continuum is divided into a triangular finite element mesh, and a 4-node joint element with bonding effect is inserted on the common edge of adjacent triangular elements. Thus, the triangular elements do not share nodes, as shown in Fig. 1. FEM is used to solve the stress and strain of the triangular element, while the discrete element method is used to address the contact between triangular elements. Thus, the sliding and rotation of the block can be simulated. In addition, the fragmentation and fracture of the continuum can be modeled through joint element breaking. The fracture criterion of the joint element in FDEM is shown in Fig. 2. The deformation of the continuum can be simulated by the deformable triangular element and the unbroken joint element with a bonding effect. Next, we introduce the dynamic equations for the system, contact detection, and contact force calculation.

2.1. Dynamic equations

After the continuum is discretized, it is composed of triangular elements, joint elements and nodes. The displacement calculation of the continuum is attributed to the calculation of nodal displacement. The contact force, the reaction force induced by deformation of triangular element, and the bonding force of joint element are all assigned to the nodes. Finally, the motion equation with node displacement as unknown is constituted as

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} = \mathbf{F}_c(\mathbf{x}) + \mathbf{F}_e(\mathbf{x}) + \mathbf{F}_j(\mathbf{x}) + \mathbf{F}_{ext}(\mathbf{x}), \quad (1)$$

where $\mathbf{M}$ is the mass matrix and $\mathbf{C}$ is the damping matrix, both of which are diagonal matrices, $\mathbf{x}$ is the displacement vector of nodes. The kinetic energy of the system is consumed by the damping, by which the static problem can be solved using dynamic relaxation. The damping matrix is given by

$$\mathbf{C} = \mu \mathbf{I}, \quad (2)$$

where $\mathbf{I}$ is a unit matrix, μ is the damping coefficient. According to the single-degree-of-freedom point spring system, the critical damping coefficient is given by [2]

$$\mu = 2h\sqrt{\rho E}. \quad (3)$$

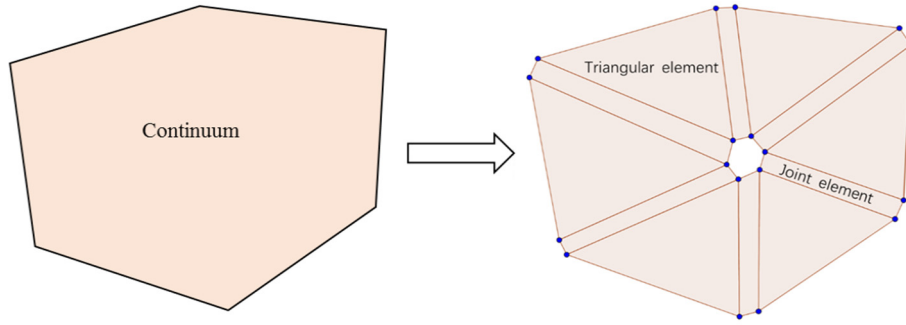


Fig. 1. Continuum characterization in 2D FDEM.

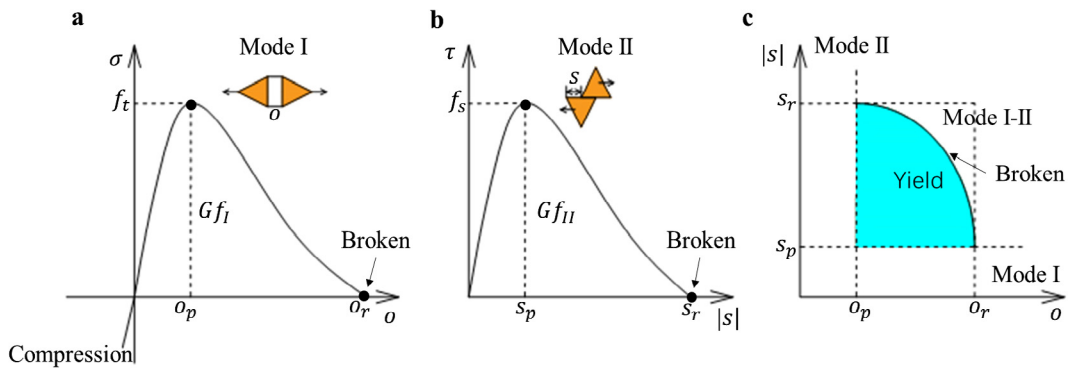


Fig. 2. Constitutive behavior of a joint element in 2D FDEM. (a) Relationship between normal bonding stress and normal opening amount in mode I; (b) Relationship between tangential bonding stress and tangential slipping amount in mode II; (c) Relationship between joint element yielded, failure and the normal opening amount and tangential slipping amount in mixed mode I-II [52,55].

For Eq. (1), using the explicit integration strategy, the velocity and displacement of each node at next time step are given by

$$v_i^{(t+\Delta t)} = v_i^{(t)} + \sum F_i^{(t)} \frac{\Delta t}{m_n} x_i^{(t+\Delta t)} = x_i^{(t)} + v_i^{(t)} \Delta t, \quad (4)$$

where $F_i^{(t)}$ is the total node force, Δt is the time step, m_n is node mass, which is equal to one third mass of the triangular element.

2.2. Contact detection and contact force calculation

In FDEM, the NBS contact detection algorithm is used to determine which triangular elements are in contact with each other. The total time taken to detect all contacts of the algorithm is proportional to the total number of elements. After finding the triangular

element contact pair, the contact force can be calculated. The contact force consists of two parts, i.e. the normal contact force and the tangential contact force. The normal contact force is calculated according to the penalty function method, while the tangential contact force is calculated according to the coulomb friction.

For a contact pair as shown in Fig. 3, the normal contact force between the two triangular elements is given by

$$\mathbf{f}_c = \int_{\Gamma_{\beta_t \cap \beta_c}} [\mathbf{grad} \varphi_c(\mathbf{P}_c) - \mathbf{grad} \varphi_t(\mathbf{P}_t)] dA, \quad (5)$$

where $\mathbf{P}_c$ is the overlapping points belonging to contactor β_c , $\mathbf{P}_t$ is the overlapping points belonging to target β_t , φ is the potential of a point in a triangular element.

$$\varphi(\mathbf{P}) = p_n \min \left\{ \frac{3A_1}{A}, \frac{3A_2}{A}, \frac{3A_3}{A} \right\}, \quad (6)$$

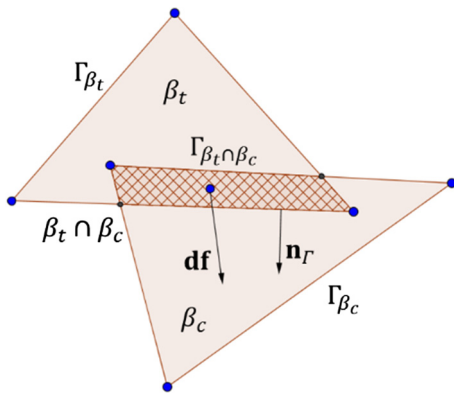


Fig. 3. The schematic diagram of the normal contact force calculation.

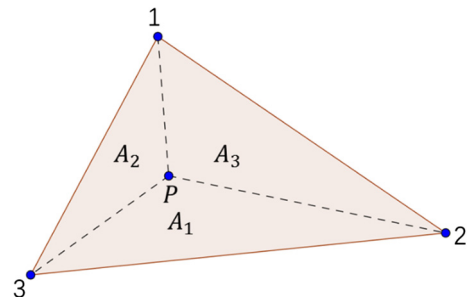


Fig. 4. The potential of a point P in a triangular element.

where p_n is the normal penalty parameter, $A_i (i = 1, 2, 3)$ is the area of corresponding sub-triangles, as shown in Fig. 4, A is the area of a triangular element.

The tangential contact force is calculated based on the coulomb friction

$$f_t^{t+1} = f_t^t - p_t \Delta u_t, \quad (7)$$

where f_t^{t+1} and f_t^t are the tangential contact force at the next time step and the current time step, p_t is the tangential penalty parameter, Δu_t is relative displacement vector of contact point at one time step.

If the tangential contact force calculated by Eq. (7) is greater than the maximum static friction, i.e. $|f_t^{t+1}| > |\mu f_n|$, then the two triangular elements in contact will be in relatively sliding and the tangential contact force is given by

$$f_t^{t+1} = -\mu f_n, \quad (8)$$

where μ is the friction coefficient, f_n is the normal contact force.

2.3. Stress calculation of the triangular element

The stress-strain relationship of the triangular element satisfy the following relationship

$$\mathbf{T} = \frac{1}{\sqrt{|\det \mathbf{F}|}} \left[\frac{E}{1+\nu} \mathbf{E}_d + \frac{E}{1-\nu} \mathbf{E}_s + 2\mu \mathbf{D} \right], \quad (9)$$

where $\mathbf{T}$ is the stress tensor, $\mathbf{F}$ is the deformation gradient, E is the elastic modulus, ν is the Poisson's ratio, $\mathbf{E}_d$ is the Green strain tensor, $\mathbf{E}_s$ is the St. Venant strain tensor, μ is the damping coefficient, $\mathbf{D}$ is the strain rate tensor, where $\mathbf{F}$, $\mathbf{E}_d$, $\mathbf{E}_s$, $\mathbf{D}$ can be calculated based on the current coordinates, the initial coordinates, the current velocity and the initial velocity of three nodes of a triangular element.

The fully coupled hydro-mechanical model in this paper exploits the advantages of FDEM and uses the unique topological connection between the joint element and the triangular element in FDEM to construct a real pore seepage model and a fracture seepage model. Combined with the mechanics fracture calculation in FDEM, the fully coupled model can simulate the deformation and fracturing of porous medium (such as rock, soil) driven by fluid.

3. The conceptual model of rock mass and basic assumptions of the fully coupled model

3.1. Conceptual model of rock mass

A rock mass is composed of joints and rock blocks cut by joints. There may be fluid in the joints or fractures. A rock block (rock matrix) contain pores and may also be partially or fully filled with fluid. Thus, the conceptual model of the rock mass as shown in Fig. 5 can be established. The cyan-filled oval zone in Fig. 5 represents the pore, and the cyan line represents the joint or fracture that may be filled by fluid. Thus, the permeability of the rock mass is composed of two parts, the permeability of the rock matrix and the permeability of the joint. The permeability of the fracture is mainly related to the aperture of the fracture, whereas the permeability of the rock matrix is related to the porosity and the stress state.

3.2. Basic assumptions

The fully coupled hydro-mechanical model proposed in this paper is based on the following assumptions: (1) the permeability of the rock matrix is characterized by the pore seepage in the triangular element, which obeys Darcy's law; (2) the permeability of fractures or joint is characterized by the fracture seepage in a parallel plate constituted by the broken joint element, which obeys the cubic law; (3) unbroken joint elements are not involved in the seepage calculation.

As shown in Fig. 1, the triangular elements and joint elements in FDEM form a unique topological connection. The fluid can flow in the triangular element and the broken joint element (i.e., fracture). The permeability of the rock matrix is taken into account by the fluid flow in the triangular element, while the permeability of fractures is characterized by fluid flow in the broken joint element. When a joint element is broken, the joint element becomes the channel of the fracture seepage and participate in the fracture seepage calculation.

4. 2D coupled hydro-mechanical model with real pore seepage and fracture seepage

4.1. The basic idea of seepage calculation and hydro-mechanical coupling

The fluid flow in the rock contains the seepage in the fracture and the pore seepage in the rock matrix, and the fluid exchange

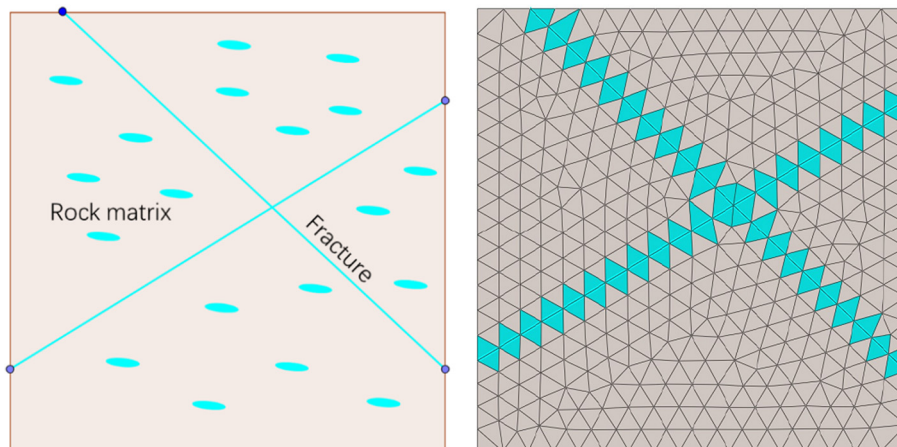


Fig. 5. Conceptual model of rock mass and the corresponding mesh in FDEM.

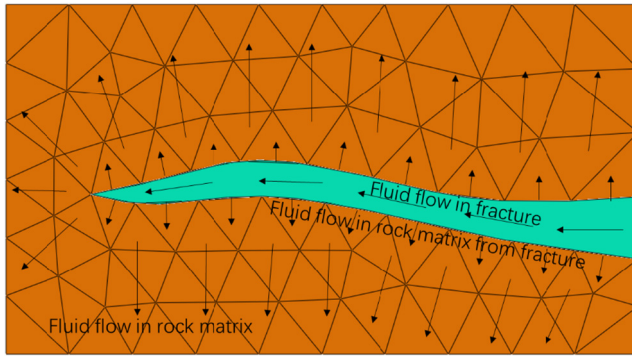


Fig. 6. Basic idea of seepage calculation.

between fracture seepage and the pore seepage occurs at the fracture, as shown in Fig. 6. Thus, the seepage calculation includes the three parts, i.e. fracture seepage, pore seepage, and the fluid exchange at the fracture between fracture seepage and porous seepage. The calculation of fracture seepage is described in Section 4.2 Fracture seepage model. The pore seepage in the rock matrix is described in Section 4.3 Two-porous porous seepage model. The fluid exchange at the fracture is described in Section 4.4 Coordinate treatment of fracture seepage and porous seepage.

Since the fracture seepage exerts a fluid pressure on the fracture surface and the pore pressure is applied to the rock matrix, the hydro-mechanical coupling model in this paper includes fracture seepage - stress coupling and pore seepage-stress coupling.

- (1) Fracture seepage-stress coupling: The fluid flow in the fracture will exert the corresponding fluid pressure on the rock wall of the fracture, and the fracture will open or close under the effect of the fluid pressure and other load, as shown in Fig. 7(a). According to the cubic law, the flow rate in the fracture is proportional to the cube of the aperture of the fracture. Thus, the fracture opening or closing will affect the fluid pressure and flow rate.
- (2) Pore seepage-stress coupling: The fluid flow in the rock matrix will impose a corresponding pore pressure on the rock matrix, as shown in Fig. 7(b). Under the effect of pore pressure and other external loads, the pore volume will change and thus alter the permeability of the rock matrix. According to Darcy's law, the permeability changes of the rock matrix will affect the fluid pressure and fluid flow in the matrix.

4.2. Fracture seepage model

A model of the fracture seepage calculation is shown in Fig. 8, which displays a part of the hydraulic fracture network in the FDEM. The broken joint elements with green color form a flow path of the hydraulic fracture network (such as the joint element between pressure node 1 and pressure node 2 in Fig. 8), while the other unbroken joint elements will not allow fluid flow.

The solution procedure of the flow between the two pressure nodes is as follows. First, the total pressure difference between the two pressure nodes is calculated. Taking pressure node 1 as an example, the total pressure difference between node 2 and node 1 is given by

$$\Delta p = p_2 - p_1 + \rho_w g(y_2 - y_1), \quad (10)$$

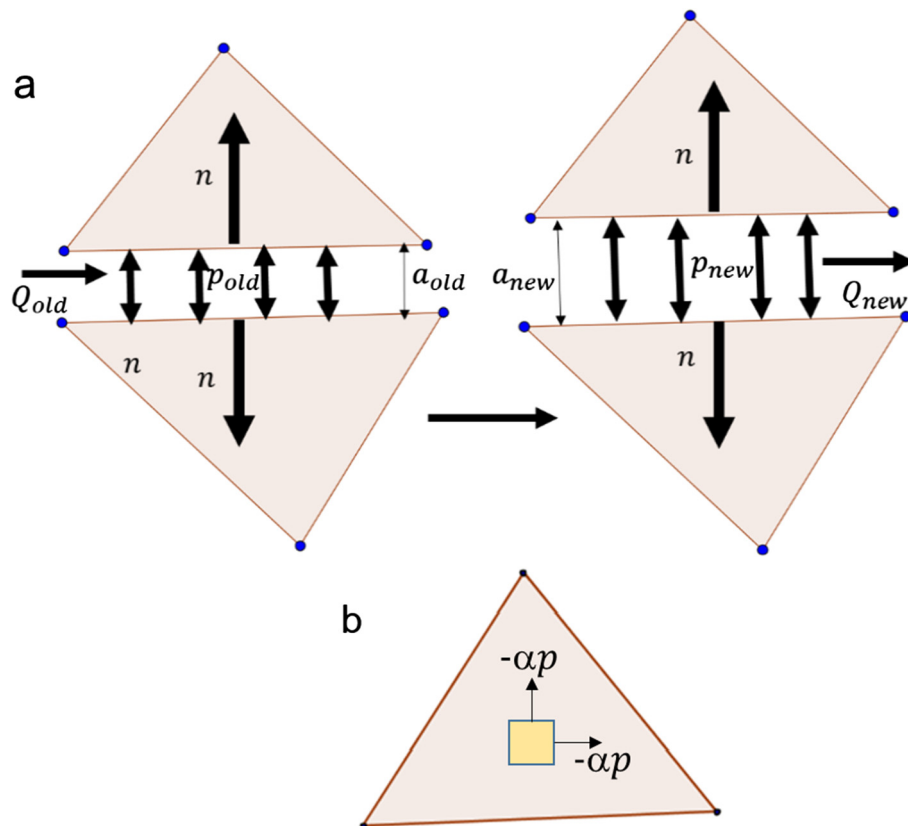


Fig. 7. Basic principle of seepage-stress coupling in this fully coupled model. (a) Fracture seepage-stress coupling: the effect of fluid pressure on fractures and the effect of fracture opening or closing on fluid pressure and velocity. (b) Pore seepage-stress coupling: pore pressure to the rock matrix.

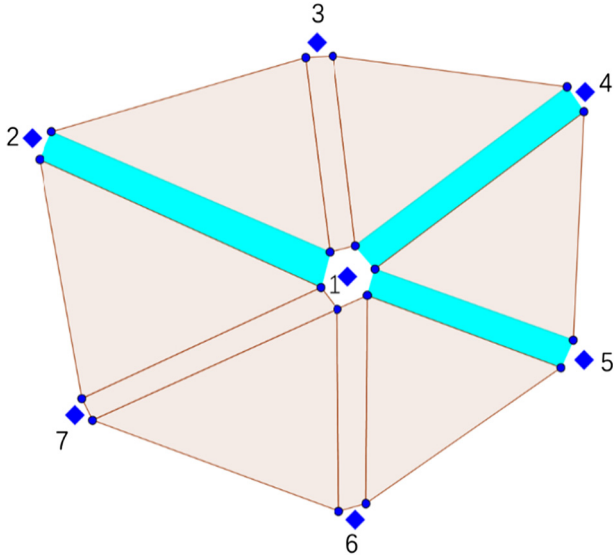


Fig. 8. Fluid flow in a hydraulic fracture network.

where p_1 and p_2 are the pressure at pressure node 1 and node 2, respectively.

Then, according to the cubic law, the flow rate between node 2 and node 1 can be obtained as

$$q_{21} = \frac{1}{12\mu} a^3 \frac{\Delta p}{L}, \quad (11)$$

where μ is the dynamic viscous coefficient of the fluid, L is the distance between the two pressure nodes, and a is the average aperture of the broken joint element, which depends on the average normal opening amount of the broken joint element u_n as shown in Fig. 9. Since the average aperture a cannot be less than or equal to zero in the seepage calculation, we set a minimum hydraulic aperture a_{res} . The average normal opening amount of the broken joint element u_n can be obtained from the relative displacement of the two end points of the joint element, i.e. $u_n = \frac{1}{2}(\mathbf{u}_i + \mathbf{u}_j) \cdot \mathbf{n}$, $\mathbf{n}$ is the normal vector of the joint element, $\mathbf{u}_i$ and $\mathbf{u}_j$ are the relative displacement vectors at the two endpoints of the joint element, respectively, as shown in Fig. 10.

Since pressure node 1 is also connected to pressure nodes 4 and 5, the flow rate q_{41} , q_{51} between pressure node 1 and pressure nodes 4, 5 can be obtained in the same way. Thus, the flow rate at pressure node 1 is $Q = q_{21} + q_{41} + q_{51}$.

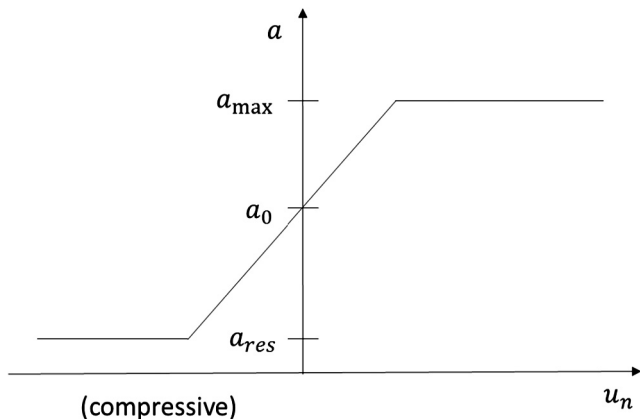


Fig. 9. Relationship between the hydraulic aperture and the normal opening amount of the broken joint element.

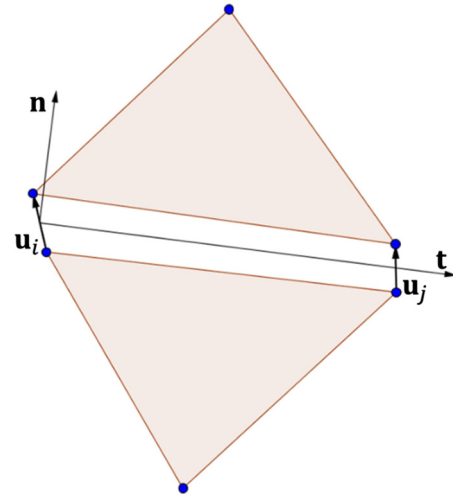


Fig. 10. Average normal opening amount of a broken joint element.

Then, the current pressure of pressure node 1 can be calculated as follows [56]

$$p_{t+\Delta t}^f = p_t^f + K_w Q \frac{\Delta t}{V} - K_w \frac{\Delta V}{V_m}, \quad (12)$$

where p_t^f is the fluid pressure of pressure node 1 at the previous step, K_w is the bulk modulus of the fluid, Q is the total flow at pressure node 1 (the outflow is negative while the inflow is positive), Δt is the time step, and $\Delta V = V - V_0$, $V_m = \frac{(V+V_0)}{2}$, where V and V_0 are half the volume of all unbroken elements connected to pressure node 1 at the previous time step and current time step, respectively.

It should be pointed out that the seepage calculation of this paper is based on the explicit integration strategy to update the pressure of the pressure node, the critical time step is given by

$$\Delta t_f = \min \left[\frac{V}{K_w \sum_i \frac{a_i^2}{12\mu}} \right] \quad \text{裂隙渗流时间步} \quad (13)$$

If the average aperture a take a larger value, the time step will be very small resulting in extremely time-consuming calculations. Thus, we set a maximum aperture a_{max} to prevent the time step too small. when $a > a_{max}$, we set $a = a_{max}$, as shown in Fig. 9.

After calculating the fluid pressure at each pressure node, the fluid pressure is applied linearly on the two boundaries of the triangular elements connected by the joint element. As with the contact force and other external loads, the surface load is converted into nodal forces and applied to the corresponding nodes of the triangular elements.

4.3. Two-dimensional porous seepage model in rock matrix

According to Darcy's law, the flow rate per unit cross-sectional area along direction i can be expressed as

$$q_i = -\frac{k_{ij}}{\mu} \frac{\partial (p - \rho_w g_k x_k)}{\partial x_j}, \quad (14)$$

where q_i is the flow rate along direction i , k_{ij} is the intrinsic permeability tensor, μ is the viscosity coefficient of the fluid, p is the fluid pressure, ρ_w is the fluid density, g_i ($i = 1, 2$) are the two components of the gravitational acceleration, and g is the modulus of the gravitational acceleration.

For convenience in the subsequent discussion, we define the quantity ϕ as the total head

$$\phi = (P - \rho_w x_k g_k) / (\rho_w g). \quad (15)$$

Thus, according to the Eq. (15), Eq. (14) can be rewritten as

$$q_i = -\rho_w g \frac{k_{ij}}{\mu} \frac{\partial \phi}{\partial x_j}. \quad (16)$$

For any given volume V , the change in fluid pressure can be given by [57]

$$\frac{\partial p}{\partial t} = \frac{M}{V} \left(Q_{total} + \alpha \frac{\partial V}{\partial t} \right), \quad (17)$$

where Q_{total} is the net flow into volume V per unit time, M is the Biot modulus, and α is the Biot coefficient.

Since the mesh shown in Fig. 1 is used in FDEM, a pore seepage model is constructed as shown in Fig. 11. As shown in Fig. 11, one third of the mass of all triangular elements connected to a node is equivalent to the node, and pore pressure of the node represents the pressure of the hexagonal zone around the node. Thus, the pore pressure distribution of the porous medium can be characterized by these discrete nodes. In the following part, we describe how to obtain the pore pressure field in the continuum based on the topological connections shown in Fig. 11.

Taking node 1 in Fig. 11 as an example, there are 6 triangular elements connected to node 1, including $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 167$ and $\Delta 172$. We refer to the hexagonal zone surrounding node 1 as a domain. The pore seepage calculation of this domain is equivalent to node 1, and the pore pressure of the whole hexagonal zone is represented by the pore pressure of node 1. Since the pore pressure at nodes 2, 3, 4, 5, 6, 7 directly connected to node 1 may be different from the pressure at node 1, pore seepage may occur in these triangular elements. For one triangular element $\Delta 123$ connected to node 1 as an example, assuming that the total head in the triangular element follows a linear distribution, the total head gradient in the triangular element is constant and can be expressed as

$$\frac{\partial \phi}{\partial x_i} = \frac{1}{A} \int_A \frac{\partial \phi}{\partial x_i} dA. \quad (18)$$

According to the Gaussian divergence theorem, Eq. (18) can be written as

$$\frac{\partial \phi}{\partial x_i} = \frac{1}{A} \int_s \phi n_i ds = \frac{1}{A} \sum_{m=1}^3 \bar{\phi}^m \in_{ij} \Delta x_j^m \quad (19)$$

where A is the area of the triangular element, n_i is the outer normal unit vector, $\bar{\phi}^m$ is the average pore pressure at the edge m , Δx_j^m is the

difference between the coordinate components of the two vertices of the edge m , and $\in_{ij}$ is the two-dimensional permutation tensor

$$\in = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus, substituting Eq. (19) into Eq. (16), the flow rate along direction i can be obtained as

$$q_i = -\rho_w g \frac{k_{ij}}{\mu} \left(\frac{1}{A} \sum_{m=1}^3 \bar{\phi}^m \in_{jk} \Delta x_k^m \right). \quad (20)$$

It should be noted that, according to Eq. (14), the flow rate in the triangular element is not zero under the effect of gravity when the fluid pressure at the three nodes of the triangular element are all zero. This result is unreasonable because the fluid should not flow out from the zone where the saturation is zero. In addition, the flow rate between the two nodes should decrease with decreasing saturation. Thus, Eq. (20) should be multiplied by a coefficient f_s , which is a function of the average saturation $\bar{s}$:

$$f_s = \bar{s}^2 (3 - 2\bar{s}), \quad (21)$$

where saturation $\bar{s}$ is the average saturation of the outflow node. The empirical function has the following properties: if $\bar{s} = 0$, $f_s = 0$; if $\bar{s} = 1$, $f_s = 1$. In addition, the derivative of the function f_s at $\bar{s} = 0$, 1 is zero. Therefore, if Eq. (20) is multiplied by the coefficient, flow will not occur in the zone in which the saturation is zero. Thus, the final flow rate along direction i can be expressed as

$$q_i = -\rho_w g \frac{k_{ij}}{\mu} \frac{\partial \phi}{\partial x_j} f_s. \quad (22)$$

Then, the flow into node 1 per unit of time can be calculated by

$$Q^n = -\frac{q_i n_i^{(n)} S^{(n)}}{3}, \quad (23)$$

where $n_i^{(n)}$ is the outer normal unit vector of the face opposite node n .

Thus, we obtain the flow $Q_{\Delta 123}$ into node 1 from the triangular element $\Delta 123$. Similarly, we can obtain the flow into node 1 from other triangular elements that directly connect to node 1. Finally, the total flow into node 1 per unit time can be expressed as

$$Q_{total} = Q_{\Delta 123} + Q_{\Delta 134} + Q_{\Delta 145} + Q_{\Delta 156} + Q_{\Delta 167} + Q_{\Delta 172}. \quad (24)$$

Then, according to the finite difference form of Eq. (17), the pore pressure of node 1 at the next time step can be given by

$$p_{t+\Delta t}^p = p_t^p + \frac{M(Q_{total} - \Delta V_{mech})}{V} \Delta t, \quad (25)$$

where p_t^p and $p_{t+\Delta t}^p$ are the pore pressure at the current time step and the next time step.

Similarly, the pore pressure of other discrete nodes can be obtained according to the above-described procedure. Then, we can obtain the pore pressure evolution of the entire solution domain.

The time step must be less than the critical time to ensure numerical stability because the pore seepage calculation in this paper adopts an explicit algorithm. The critical time step is given by

$$\Delta t_f = \min \left[\frac{V}{M \sum_i k_i / \mu} \right], \quad \text{孔隙渗流时间步} \quad (26)$$

where k_i is the isotropic intrinsic permeability.

4.4. Coordinate treatment of fracture seepage and porous seepage

Since both the pore seepage and the fracture seepage are taken into account, there is an interaction between the fracture seepage

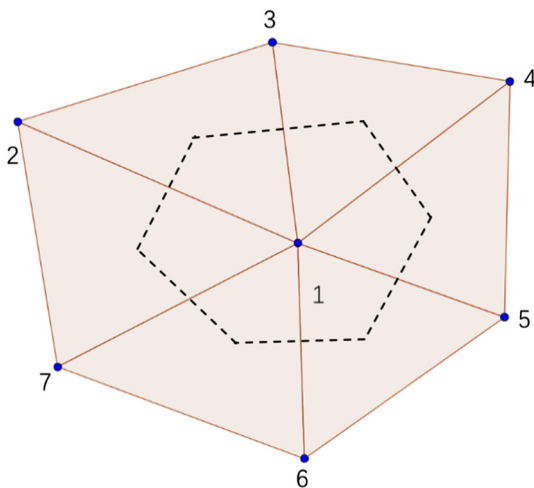


Fig. 11. The pore seepage model in the fully coupled model.

$$\sigma'_{ij} = \sigma_{ij} - \alpha p \delta_{ij}, \quad (33)$$

where α is the Biot coefficient.

Therefore, the pore pressure will cause the stress field in the rock matrix to change, and the stress increment due to the pore pressure is given by

$$\Delta \sigma_{ij} = -\alpha p \delta_{ij}. \quad (34)$$

In this way, we obtain the stress increment caused by the pore pressure, which will be applied to the triangular element as a volume force.

4.6. Fracture-stress-seepage coupling process

Fracture seepage will interact with the rock matrix, and pore seepage will cause a stress change in the rock matrix. Therefore, the hydro-mechanical coupling includes fracture seepage-stress coupling and pore seepage-stress coupling.

4.6.1. Fracture initiation and propagation

In FDEM, the continuum is divided into a triangular finite element mesh, and the joint element is inserted between the triangular elements. The coupled model in this paper simulates crack initiation, propagation and intersection in rock via joint element breaking. Under the action of the fluid pressure and other loads, the joint element will break (i.e., new fractures will be generated). These newly generated fractures and the natural fractures existing in the rock mass constitute the network channel of the fracture seepage.

4.6.2. The effect of fracture seepage on the stress field

The fracture flow in the rock mass has fluid pressure and tangential viscous force on the two walls of a fracture. The effect of the fluid pressure and the tangential viscosity force of the fluid is shown in Fig. 13.

4.6.3. The effect of stress variation and joint element breaking (i.e., fracture generation) on fracture seepage

The stress change will affect the aperture of the broken joint element (i.e., fracture), which will change the permeability of the joint element. When the aperture of the broken joint element changes, according to Eq. (11), both the flow rate and the fluid pressure in the fracture will change.

The breaking of the joint element changes the network channel of the fracture seepage, which will alter the flow velocity and fluid pressure in the fracture seepage, i.e. the fracture seepage field is changed. Thus, combination with the joint element breaking and Eq. (11) will allow the fully coupled model to handle this effect of fracture generation on fracture seepage automatically. Other seepage-stress coupling models that do not consider crack initiation and propagation cannot deal with this effect.

4.6.4. The effect of pore seepage on the stress field

The pore seepage in the rock matrix will exert a volume force on the rock matrix. According to the effective stress principle, the volume force is given by Eq. (34).

The effect of pore pressure is also required for the unbroken joints element (i.e., the fluid force as in Eq. (34) is also applied to the two edges of the triangular element connected by the unbroken joint element). This force can be transformed into the normal pressure on both edges, similar to that shown in Fig. 13.

4.6.5. The effect of stress variation on pore seepage

The stress variation in the rock matrix will cause the pore size to change, while the change in the pore size will alter the perme-

ability of rock matrix. That is, the stress in the rock matrix will influence the permeability of the rock matrix.

Here, we use the following coupling equation to reflect the influence of stress on the permeability [58]

$$(\sigma, p) = \xi k_0 e^{-\beta(\sigma_{ii}/3 - \alpha p)} \sigma_{ii} = \sigma_{11} + \sigma_{22} + \sigma_{33} \quad (35)$$

where ξ ($\xi \geq 1$) is a mutation coefficient of permeability that accounts for the increase in permeability once the element reaches the damage state in [58]. In this paper, since we do not consider the damage of the triangular element, we set $\xi = 1$; $\sigma_{ii}/3$ is the average total stress, β is the coupling parameter that reflects the effect of stress on the permeability of rock matrix, α is the Biot coefficient, and p is the pore pressure.

5. Examples

5.1. Example 1: Unsteady saturated pore seepage

As shown in Fig. 14, the model is 1 m in length and 0.1 m in width. The model is divided into 2000 triangular elements, each with a size of 0.01 m. The porosity of the model is 0.1, and the intrinsic permeability is $1 \times 10^{-11} \text{ m}^2$. The left and right boundaries are subject to constant pressure; the fluid pressure is 100 kPa at the left boundary and 0 kPa at the right boundary fluid pressure. The initial saturation of the model is 1.0. The density of the fluid ρ_w is 1000 kg/m³, the viscosity coefficient μ is 0.001 Pa s, and the bulk modulus of fluid K_w is 2.2 GPa. The change in the pore pressure over time in the model was studied.

The analytical solution of this problem is given by

$$P(x, t) = P_1 + \frac{x}{L}(P_2 - P_1) + \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-\kappa n^2 \pi^2 t / L^2} \left(\frac{P_2 \cos(n\pi) - P_1}{n} \right) \sin \frac{n\pi x}{L}, \quad (36)$$

where P_1 is the fluid pressure at the left boundary; P_2 is the fluid pressure at the right boundary; L is the length of the model; t is time; x is the distance from the left boundary; and $\kappa = (k/\mu)M$, where k is the intrinsic permeability, μ is the viscosity coefficient of porous media, and M is the **Biot modulus, here $M = K_w/n$** .

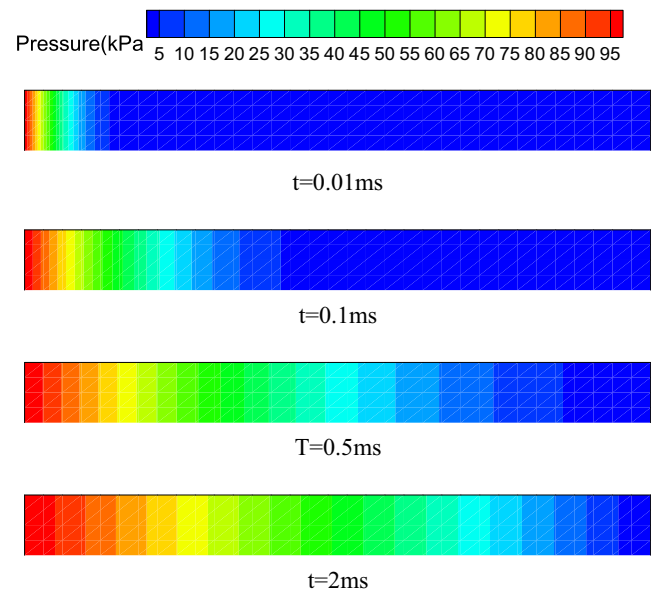


Fig. 14. Pore pressure distribution at different times.

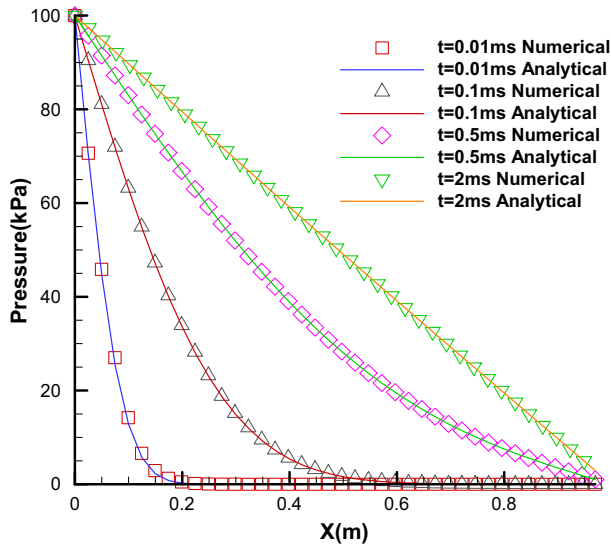


Fig. 15. Numerical and analytical solutions of pore pressure at different times.

The pore pressure distribution at times of $t = 0.01$ ms, 0.1 ms, 0.5 ms and 2 ms is shown in Fig. 15. The numerical solutions agree well with the analytical solutions. The results demonstrate that the proposed model correctly deals with the problem of saturated unsteady flow. Since the steady-state seepage is the final state of the unsteady seepage, the proposed algorithm also correctly deals with steady-state seepage. For example, at $t = 2$ ms, the pore seepage in the model reaches a steady state, and the pore pressure in the model is linearly distributed.

5.2. Example 2: Unsteady unsaturated pore seepage

In the two models shown in Fig. 16(a) and (b), the saturation of the model is zero at the beginning. The upper and lower boundaries of the model in Fig. 16(a) are impervious, and the left boundary still exerts constant fluid pressure, whereas the right boundary does not impose any constraints. In Fig. 16(b), the model in Fig. 16

(a) is placed in the vertical position (i.e., the fluid flow is affected by the effect of gravity), and the boundary conditions are unchanged. The position of the wet and dry dividing line with time is determined under the effect of pore water pressure in the model in Fig. 16(a) and under the combined effect of pore pressure and gravity in the model in Fig. 16(b).

The analytical solutions for models in Fig. 16(a) and (b) are given by [59]

$$X(t) = \sqrt{2Pt}, \quad (37)$$

$$t = \frac{-X(t)}{\alpha} - \frac{P}{\alpha^2} \ln \left(1 - \frac{\alpha X(t)}{P} \right), \quad (38)$$

where $X(t)$ is the distance between the wet and dry dividing line and the injection boundary; $P = (kP_0)/(\mu n)$, where k is the intrinsic permeability, μ is the fluid viscosity coefficient, and n is the porosity; and $\alpha = (\rho_w g k)/(\mu n)$, where ρ_w is the fluid density and g is the gravity acceleration.

The calculated parameters are as follows: the porosity of the model n is 0.1, the intrinsic permeability k is $1 \times 10^{-9} \text{ m}^2$, and the initial saturation s is 0. The density of the fluid ρ_w is 1000 kg/m³, the fluid viscosity coefficient μ is 0.001 Pa s, the Biot modulus M is 2.2×10^7 Pa, P_0 is 1×10^4 Pa, and the gravitational acceleration g is 9.8 m/s². 水是10次方，但是为了减少计算时间，特地取值为7次

The numerical solution obtained by the fully coupled model in this paper is shown in Fig. 17. The numerical solution agrees well with the analytical solution. The correctness of the algorithm for the unsaturated seepage problem is verified.

5.3. Examples 3: Coupling of seepage-stress in porous media

Fig. 18 shows a saturated soil layer with a thickness of 20 m laid on a rigid, impermeable base [57]. Under the undrained condition, a constant surface load $p_z = 1 \times 10^5$ Pa is applied to the top boundary of the soil layer. The soil matrix is a homogeneous elastomer in which the fluid flow is described by the isotropic Darcy's law. Initially, the applied surface loads are all carried by the fluid. Then, the fluid is drained from the soil layer surface gradually, and the surface load is continually transferred from the fluid to the soil layer. The analytical solution of this one-dimensional consolidation

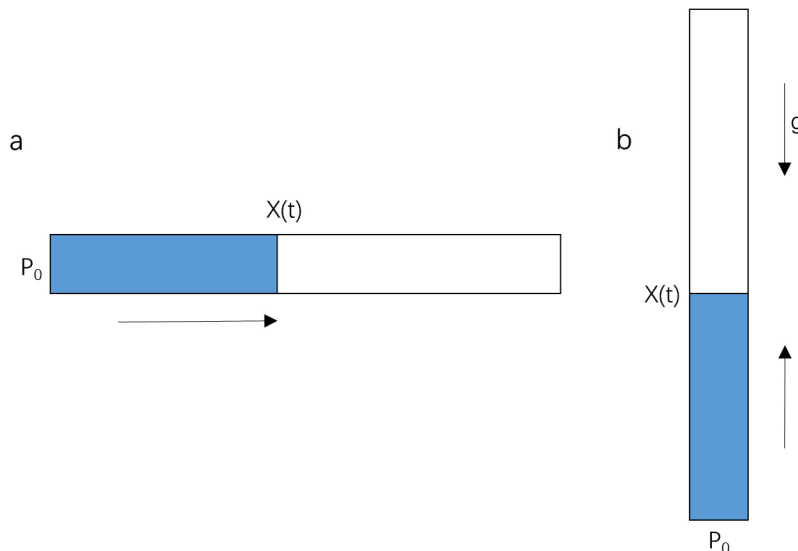


Fig. 16. Unsaturated pore seepage example [59]. (a) Fluid filled with a porous medium strip (no gravity). (b) Fluid filled with the porous medium strip under the effect of gravity.

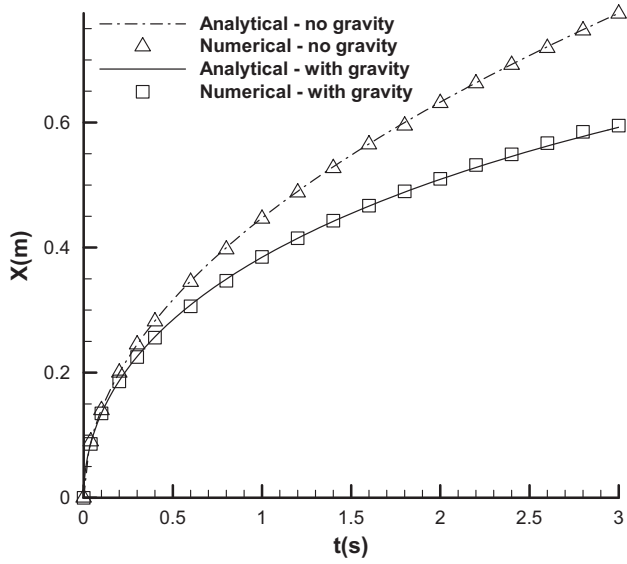


Fig. 17. Position of the wet and dry dividing lines at different times.

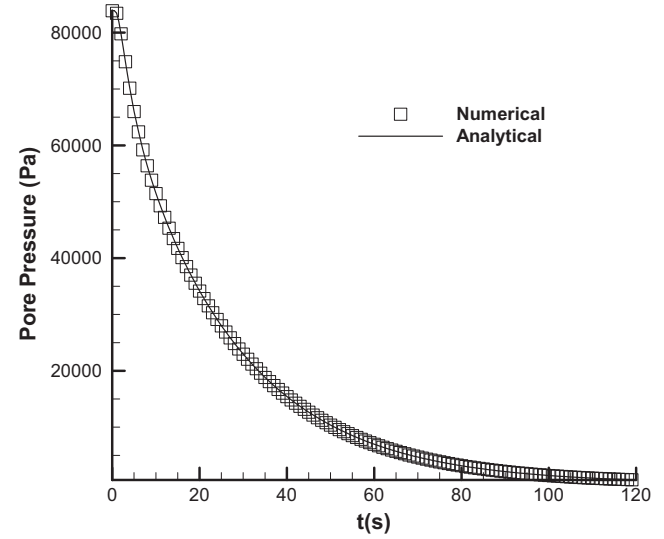


Fig. 19. The pore pressure at the midpoint of the model varies with time.

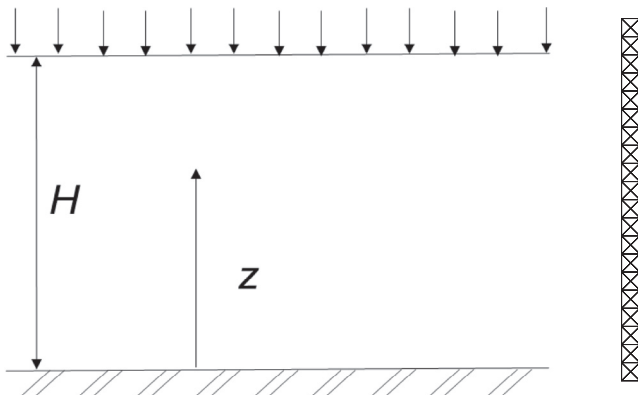


Fig. 18. One-dimensional consolidation model and computational mesh model [57].

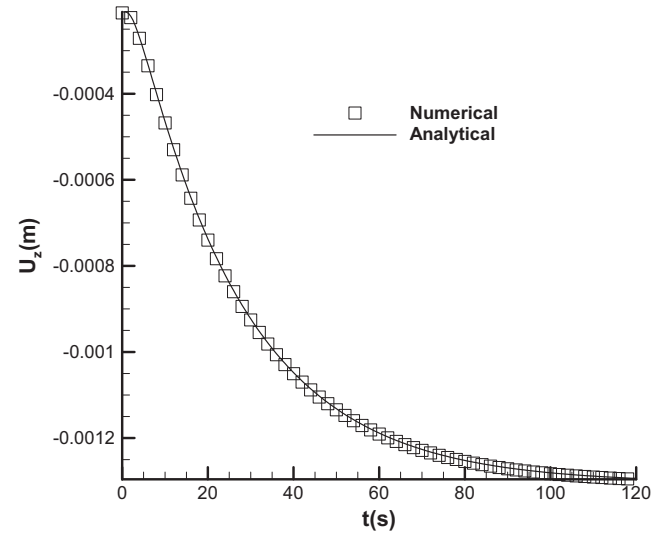


Fig. 20. The vertical displacement of the midpoint varies with time.

problem can be obtained according to Biot consolidation theory [60].

In this problem, the analytical solution of pore pressure is given by

$$p = 2p_0 \sum_{m=0}^{\infty} \frac{\sin(a_m \hat{z})}{a_m} e^{-a_m^2 \hat{t}}, \quad (39)$$

where $p_0 = \frac{\alpha}{\alpha_1} \frac{1}{S} p_z$, $a_m = \frac{\pi}{2}(2m+1)$, $\hat{z} = \frac{H-z}{H}$, and $\hat{t} = \frac{ct}{H^2}$, where $c = k/(\mu S)$, $S = \frac{1}{M} + \frac{\alpha^2}{\alpha_1}$ is the storage water coefficient, M is the Biot modulus, and α is the Biot coefficient.

The analytical solution of vertical displacement is given by

$$u_z = \frac{Hp_z}{\alpha_1} \left(2\alpha \frac{p_0}{p_z} \left[\sum_{m=0}^{\infty} \frac{\cos(a_m \hat{z})}{a_m^2} e^{-a_m^2 \hat{t}} \right] + \hat{z} - 1 \right). \quad (40)$$

The calculation mesh is shown in Fig. 18, the left, right and bottom boundaries of the model are fixed along the normal direction. The calculated parameters are as follows: the drained bulk modulus K of the porous media is 5×10^8 Pa, the shear modulus G is 2×10^8 Pa, the Biot modulus M is 4×10^9 Pa, the Biot coefficient α is 0.1, the intrinsic permeability k is 1×10^{-12} m², the density of the fluid ρ_w is 1000 kg/m³, and the viscosity coefficient of fluid μ

is 0.001 Pa s. We establish a monitoring point at $z = H/2$. The pressure and displacement of the monitoring point with time is as shown in Figs. 19 and 20, respectively. The numerical solution agrees well with the analytical solution, which verifies the correctness of the coupled model in this paper in dealing with unsteady seepage-stress coupling problems.

5.4. Hydraulic fracturing application examples

As shown in Fig. 21, there are two sets of natural fractures in the square zone. The edge length of the square zone is 50 m. The dip angle of two sets of natural fractures is -34° and 85° (counter-clockwise is positive). In this zone, the horizontal in situ stress σ_h is 20 MPa, and the vertical in situ stress σ_v is 10 MPa. There is an injection point at the center of the square zone. The fluid pressure at this point is fixed at 20 MPa. We use the coupled model in this paper to simulate the problem and study the interaction between the newly generated hydraulic fracture and the natural fracture.

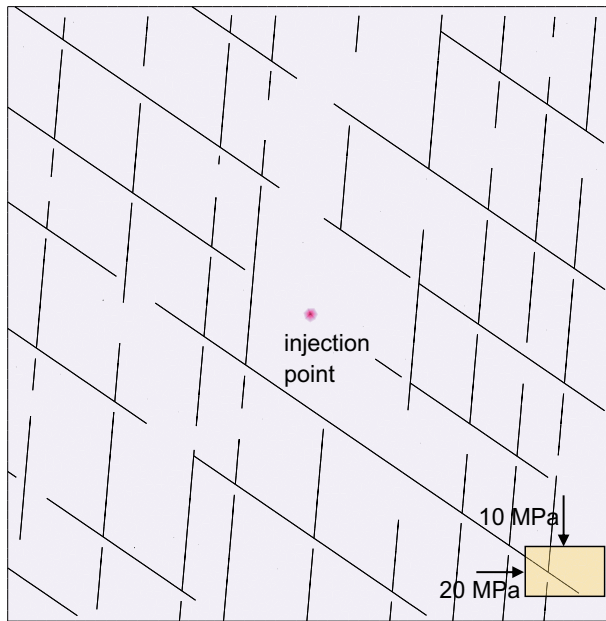


Fig. 21. Geometric model.

Table 1
Mechanical input parameters.

Parameters	Value
Rock	
Bulk density, ρ (kg/m ³)	2300
Young's modulus, E (GPa)	17
Poisson's ratio, ν	0.29
Joint element	
Tensile strength, f_t (MPa)	5
Internal cohesion, c (MPa)	20
Friction angle of intact material, ϕ (°)	35
Friction angle of fractures, ϕ (°)	35
Fracture energy release rate, G_f (J/m ²)	2.0
Fracture energy release rate, G_{fII} (J/m ²)	10.0
Normal contact penalty, p_n (GPa)	1000
Tangential contact penalty, p_t (GPa/m)	1000
Fracture penalty, p_f (GPa)	1000
Initial aperture a_0 (m)	1×10^{-6}
Minimum aperture a_{res} (m)	1×10^{-6}
Maximum aperture a_{max} (m)	3×10^{-4}

The input mechanics parameters are shown in Table 1. The seepage calculation parameters are as follows: the density of fluid ρ_w is 1000 kg/m³, the viscosity coefficient of fluid μ is 0.001 Pa.s, the fluid bulk modulus K_w is 2.2 GPa, the porosity of the rock matrix n is 0.1, the intrinsic permeability k is 1×10^{-12} m², and the initial aperture a_0 and minimum aperture a_{res} of the natural fracture in the two groups is 1×10^{-4} m, the maximum aperture of the natural fracture is 3×10^{-4} m. The time step is 2.5×10^{-6} s. The results simulated by the coupled model in this paper are shown in Fig. 22.

As shown in Fig. 22, the hydraulic fracture initiates from the injection point and propagates along the direction of the maximum principal stress in the in-situ stress, as shown in steps 500–1000 in Fig. 22. The stress distribution at step 1500 is shown in Fig. 23. The crack tip of the newly generated hydraulic fracture then intersects the natural fractures with an angle of 85°. The newly generated hydraulic fracture crosses the approximately vertical natural fractures and continues to propagate along the horizontal direction

(step 1500 in Fig. 22). Although encountering natural fractures, the newly generated fracture does not arrest or turn, mainly because the angle between the plane of the natural fracture and the direction of maximum principal stress in the in-situ stress is large (85°). Thus, the horizontal in situ stress (compressive stress) has a large component acting on the natural fracture to prevent the propagation of the natural fracture ends. Thus, the newly generated hydraulic fracture continues to propagate along the direction of the maximum principal stress, consistent with research results in the literature [61]. After it crosses the two natural fractures with an inclination of 85° along the horizontal direction, the left newly generated hydraulic fracture intersects the natural fracture with an inclination of -34° (step 3000 in Fig. 22). However, the newly generated hydraulic fracture is arrested by the natural fracture. A new hydraulic fracture subsequently initiates from the end of the natural fracture, as shown in step 5000 in Fig. 22. This is different from the case of a hydraulic fracture intersecting with the natural fracture with an inclination angle of 85° because the compressive stress component of the in situ stress in the natural fracture is relatively small, which makes it easier to propagate along the end of the natural fracture. Due to the control effect of in situ stress, the new hydraulic fracture does not propagate along the direction of the original natural fracture after initiation at the natural fracture end but instead continues to propagate along the horizontal direction (i.e., the direction of the maximum principal stress), as shown in steps 5000–25,000 in Fig. 22 and consistent with research results in the literature [61]. After the new hydraulic fracture on the right side intersects the natural fracture with an inclination angle of -34° , the new hydraulic fracture ceases to propagate. At the end of the natural fracture, a new hydraulic fracture initiates and continues to propagate as shown in steps 7000–35,000 in Fig. 22.

It can be concluded from the above discussion that the in-situ stress plays a decisive role in determining the direction of crack propagation. The existence of natural fractures will locally affect the propagation of hydraulic fractures, resulting in arrest of the newly generated hydraulic fracture or continued propagation along the natural fracture. When the newly generated hydraulic fracture intersects with the natural fracture at a large angle, the propagation of the newly generated hydraulic fracture may not be affected by the natural fissures and continue to propagate along the direction of the maximum principal stress.

In addition, the evolution of the fluid pressure in Fig. 22 reveals that the fluid preferentially flows along the newly generated hydraulic fracture as shown in step 1000 in Fig. 22 because the permeability of the rock matrix is very small relative to the hydraulic fractures and natural fractures. When the new hydraulic fracture intersects the natural fracture, the fluid preferentially flows into the natural fracture, as shown in steps 1500–9000 in Fig. 22. Thus, the fluid pressure distribution displays a strip along the fracture network formed by the newly generated hydraulic fracture and the natural fracture. Then, as the time step increases, the fluid in the newly generated hydraulic fractures and natural fractures gradually infiltrates the surrounding rock matrix, resulting in a wider banded distribution of the fluid pressure along the fracture network, as shown in steps 25000–140,000 in Fig. 22.

As shown in Fig. 22, the proposed coupling model can capture crack initiation, propagation, intersection, the interaction of the newly generated hydraulic fracture and the natural fracture, and the evolution of fluid pressure during the fracture process in a formation with an arbitrary complex fracture network. Conventional methods often face difficulty in dealing with fracture simulations in which the formation contains a very complex fracture network.

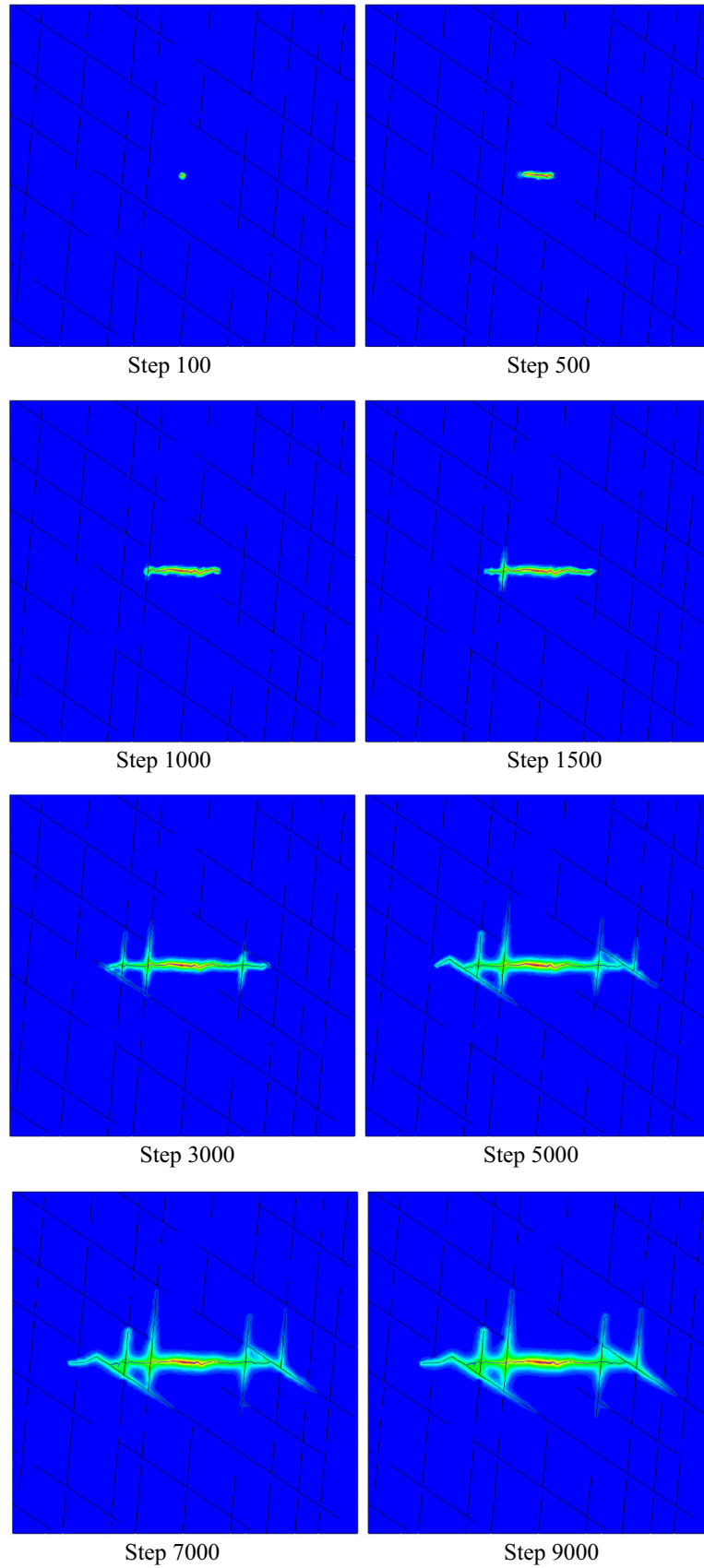


Fig. 22. The distribution of fluid pressure and crack propagation during hydraulic fracturing.

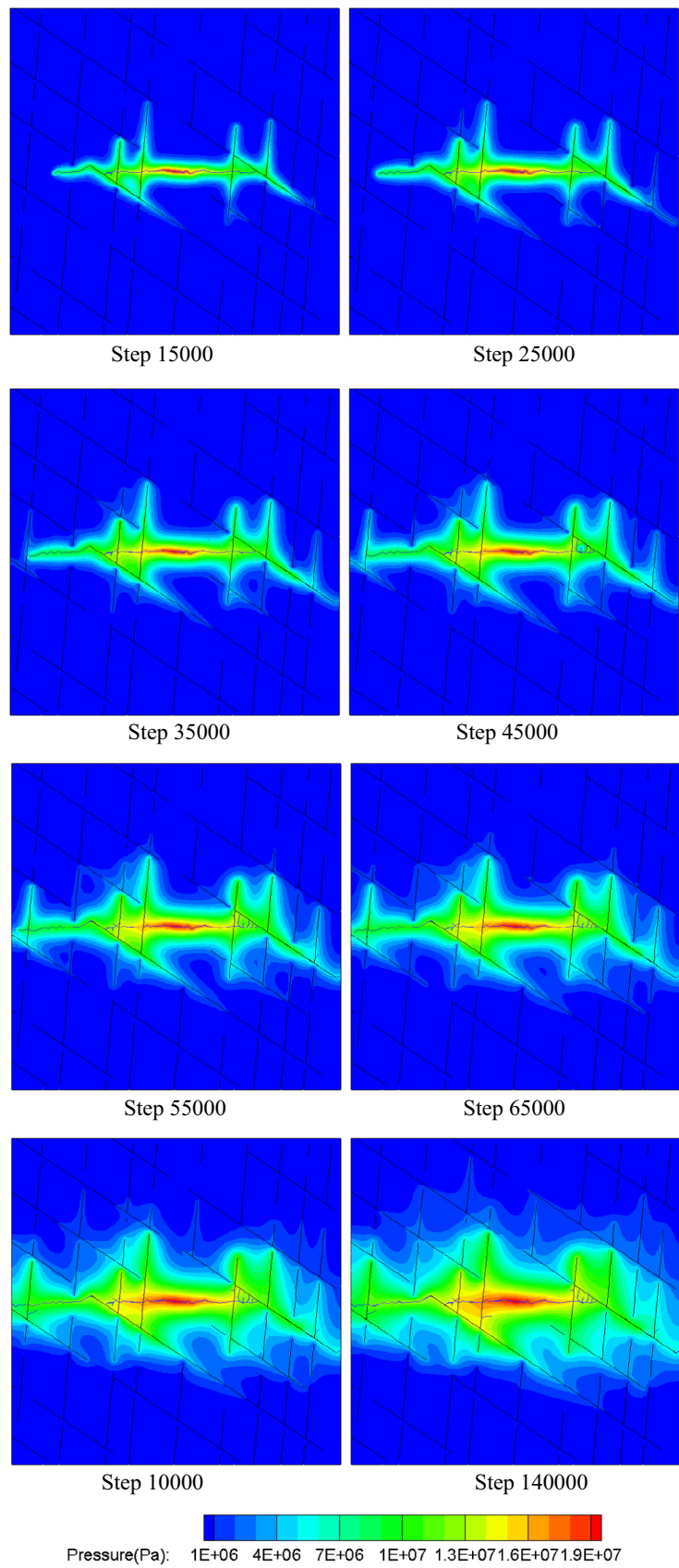


Fig. 22 (continued)

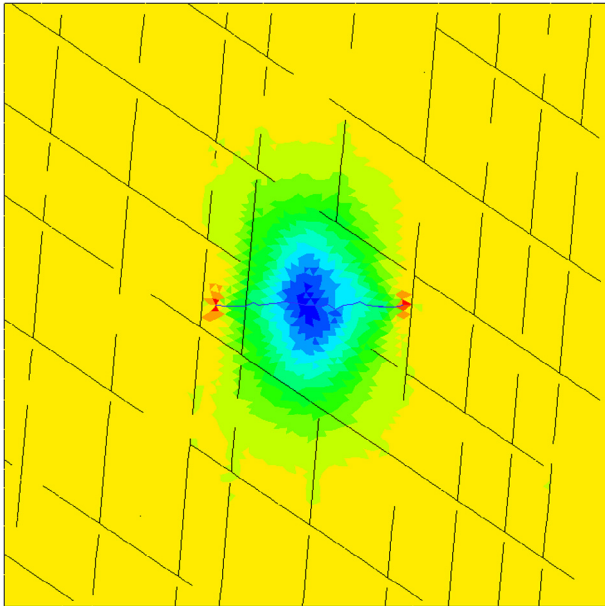


Fig. 23. The maximum principal stress distribution at step 1500.

6. Conclusion

In this paper, a two-dimensional fully coupled hydro-mechanical model for simulating the deformation and fracture of porous media driven by fluid is presented. This model calculates the fluid flow in fractures by fracture seepage, whereas the fluid flow in rock matrix is calculated by real pore seepage. In combination with the advantages of FDEM for the simulation of solid fracturing, this approach can simulate hydraulic fracturing in a formation that initially contains an arbitrary complex fracture network. It can capture crack initiation and propagation, the intersection of cracks, and the interaction of hydraulic fractures and natural fractures during the fracture process, as well as the pressure distribution in the rock mass. The proposed approach is verified in detail, and the numerical solutions agree well with the analytical solutions, thus verifying the correctness of the proposed approach.

The advantage of this approach over equivalent methods is that the fluid flow in the rock matrix is described by the real pore seepage, which avoids the following problems of the equivalent method: when simulating transient flow, the fluid flow of the equivalent method lags behind that of the real pore seepage. The equivalent method must calibrate the initial aperture of the joint element, and the macroscopic permeability in the equivalent method is also affected by the computational mesh.

In addition, compared with the conventional method, the approach in this paper has the following advantages: the present approach can simulate the hydraulic fracturing problem in an arbitrary complex fracture network but is not limited to it. For example, the present model can also be applied to slope failure analysis under the effect of rainfall or reservoir water, stability analysis of tunnel excavation associated with the effect of water, and stability analysis of dam foundations. Moreover, it can even solve soil mechanics problems related to water. Therefore, the approach presented in this paper represents a universal tool for geomechanical problems, and its application prospects are very broad. In the future, we will extend this approach to a three-dimensional scenario to enable more realistic modeling of engineering problems.

Acknowledgments

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